MTH 201 Chapter 2: Integration Techniques

2.7 Improper Integrals Worksheet

1. Evaluate $\int_{1}^{\infty} e^{-x}dx$ if possible.

2. Evaluate $\int_{-8}^{1} \frac{1}{\sqrt[3]{x}}dx$ if possible.

3. Evaluate $\int_{0}^{1} \ln(x)dx$ if possible.

4. Evaluate $\displaystyle\int_{-\infty}^{3} \frac{e^x}{3-2e^x}\,dx$ if possible.

5. Evaluate $\displaystyle\int_{-\infty}^{\infty} \frac{dx}{\sqrt[3]{x-1}}$ if possible.

6. Evaluate $\int_{-1}^{\infty} \frac{x^2}{\sqrt{x^3+1}}\,dx$ if possible.

7. Use the Comparison Theorem to determine whether the integral is convergent or divergent.

a) $\displaystyle\int_1^{\infty} \frac{x^4}{x^5 - 3x}\, dx$

b) $\displaystyle\int_1^{\infty} \frac{x^3 + x + 1}{x^5 + 2x + 1}\, dx$

c) $\int_{1}^{\infty} \frac{x+1}{\sqrt{x^4 - x}}\,dx$

d) $\int_{0}^{\pi} \frac{\sin^2(x)}{\sqrt{x}}\,dx$

e) $\int_0^{\infty} \frac{x}{x^3+1}\,dx$